dirac16complex: a second-quantized Pin(4,4) fermion field and the 6-planes it creates

A classroom guide for a beginning student. Everything stated here is computed, with exact arithmetic, in the Mathematica notebook 6planes.nb, which mirrors the notebook Pre-gravity6plane.nb of P. L. Nash. All equations used are written out in this document.

1. The stage: eight coordinates and a metric with four times

The coordinates are $x_0, x_1, \ldots, x_7$: x_0 is the hidden space direction, x_1, x_2, x_3 are ordinary 3-space, x_4 is time and x_5, x_6, x_7 are superluminal time directions. Lengths are measured with

$$\eta_{AB} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1), \qquad ds^2 = \eta_{AB}\, dx^A dx^B = dx_0^2+dx_1^2+dx_2^2+dx_3^2-dx_4^2-dx_5^2-dx_6^2-dx_7^2.$$

2. The eight real 16x16 Dirac matrices

With $\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$, the eight real 8×8 matrices τ_A of Pre-gravity6plane.nb ($\tau_0 = I_8$, six antisymmetric matrices built from self-dual and anti-self-dual 4×4 blocks, $\tau_7 = \tau_1 \cdots \tau_6$) and $\bar\tau_A = \sigma \tau_A^T \sigma$, the Dirac matrices are

$$T^A \equiv (T16^A)[A] = \begin{pmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{pmatrix}, \qquad T^A T^B + T^B T^A = 2\eta^{AB} I_{16}.$$

All entries are 0 or ± 1 (real). $T^0, \ldots, T^3$ are symmetric with square $+I_{16}$; $T^4, \ldots, T^7$ are antisymmetric with square $-I_{16}$. Further:

$$\Gamma = T^0T^1\cdots T^7 = \mathrm{diag}(-I_8, I_8), \qquad \sigma_{16} = T^0T^1T^2T^3 = \mathrm{diag}(-\sigma,\sigma), \qquad S^{AB} = \tfrac{1}{4}[T^A, T^B],$$

$$[S^{AB}, T^C] = \eta^{BC}T^A - \eta^{AC}T^B, \qquad (\sigma_{16}T^A)^T = -\sigma_{16}T^A, \qquad (\sigma_{16}S^{AB})^T = -\sigma_{16}S^{AB}.$$

The 28 matrices S^{AB} generate Spin(4,4); together with the reflections T^C they generate Pin(4,4).

3. The field dirac16complex

Definition. dirac16complex is a 16-component Pin(4,4) spinor field

$$\psi(x) = (\psi_0(x), \psi_1(x), \ldots, \psi_{15}(x))^T, \qquad \psi_a = \tfrac{1}{\sqrt{2}}(\xi_a + i\chi_a),$$

whose components are complex and anticommuting (Grassmann-odd): with ξ_a, χ_a real Grassmann generators,

$$\psi_a\psi_b = -\psi_b\psi_a, \qquad \psi_a\psi_b^* = -\psi_b^*\psi_a, \qquad \psi_a^2 = 0, \qquad (\theta\eta)^* = \eta^*\theta^*.$$

The last rule (complex conjugation reverses the order of factors) is the classical shadow of Hermitian conjugation of operators. dirac16complex is a second-quantized field: the ψ_a are field operators, and the Grassmann variables are their classical (path-integral) counterparts. A commuting (c-number) solution of the field equation is a mode function of the field; following Pre-gravity6plane.nb we call such a mode function the wave function of a Universe.

Dirac adjoint. $\bar\psi = \psi^\dagger\sigma_{16}$. Under Spin(4,4), $\psi \to g\psi$ with $g = \exp(\frac{1}{2}\omega_{AB}S^{AB})$, the scalar $\bar\psi\psi$ is invariant and $\bar\psi T^A\psi$ is a vector.

4. The Lagrangian: kinetic, mass (quadratic) and self-coupling (quartic) terms

$$\mathcal{L} = \tfrac{1}{2}\left(\bar\psi\, T^A \partial_A\psi - (\partial_A\bar\psi)\, T^A\psi\right) - m\,\bar\psi\psi - \frac{\lambda}{2}\,(\bar\psi\psi)^2,$$

with $m \neq 0$ and $\lambda \neq 0$ real constants and A summed over $0, \ldots, 7$. Compare with the Higgs potential $V = \mu^2|\phi|^2 + \lambda|\phi|^4$: here the mass term is quadratic and the self-coupling quartic in the spinor components; the field is a fermion, so $(\bar{\psi}\psi)^2$ is the simplest non-trivial self-interaction. Proven properties (exact Grassmann algebra, 392 monomials):

- $\mathcal{L}$ is real: $\mathcal{L}^* = \mathcal{L}$ exactly (no factor i is needed because $\sigma_{16}T^A$ is real antisymmetric).
- The mass and quartic terms are non-zero because the components are complex. For a real (Majorana) field $\xi^T\sigma_{16}\xi = 0$ identically, since σ_{16} is symmetric and $\xi_a\xi_b = -\xi_b\xi_a$.
- $\mathcal{L}$ is Spin(4,4) invariant, and invariant under the phase rotation $\psi \to e^{i\alpha}\psi$.

5. The Euler-Lagrange equations (flat space)

Treat ψ_a and ψ_a^* as independent Grassmann variables and vary ψ_a^* with left derivatives:

$$\frac{\partial^L \mathcal{L}}{\partial \psi_a^*} - \partial_\mu \frac{\partial^L \mathcal{L}}{\partial(\partial_\mu \psi_a^*)} = 0 \quad (a = 0, \ldots, 15).$$

The exact computation gives the 16 equations

$$\sigma_{16}\left(T^A\partial_A\psi - m\psi - \lambda(\bar{\psi}\psi)\psi\right) = 0 \quad \Longleftrightarrow \quad T^A\partial_A\psi = (m + \lambda\,\bar{\psi}\psi)\,\psi,$$

and varying ψ_a gives the complex-conjugate equation $(\partial_A\bar{\psi})T^A = -(m+\lambda\bar{\psi}\psi)\bar{\psi}$. Example (first component, $\lambda = 0$):

$$m\psi_4 - \partial_0\psi_{12} + \partial_1\psi_{11} - \partial_2\psi_{10} + \partial_3\psi_9 - \partial_4\psi_9 + \partial_5\psi_{10} + \partial_6\psi_{11} + \partial_7\psi_{12} = 0.$$

Conserved current. $J^A = \bar{\psi}T^A\psi$ satisfies $\partial_A J^A = \bar{\psi}(T^A\partial_A\psi) + (\partial_A\bar{\psi})T^A\psi = (m+\lambda\bar{\psi}\psi)\bar{\psi}\psi - (m+\lambda\bar{\psi}\psi)\bar{\psi}\psi = 0$.

6. Second quantization and the indefinite (Krein) metric

Quantize with x_4 as time. The momentum conjugate to ψ is $\pi = \bar{\psi}T^4$, and $\{\psi_a(x), \pi_b(y)\} = i\delta_{ab}\delta(x-y)$ gives

$$\{\psi_a(x), \psi_b^\dagger(y)\} = K_{ab}\,\delta(x-y), \qquad K = -i\,\sigma_{16}T^4, \qquad K^\dagger = K, \qquad K^2 = I_{16},$$

with eight eigenvalues $+1$ and eight -1. The space of states therefore carries an indefinite (Krein) metric. This cannot be avoided: $g_{45} = T^4T^5$ (the rotation by π in the (x_4, x_5)-plane, an element of Spin(4,4)) obeys $g_{45}^T(\sigma_{16}T^4)g_{45} = -\sigma_{16}T^4$, so it reverses the sign of the x_4-charge density $\bar{\psi}T^4\psi$; with four time directions there is no invariant notion of positive energy.

Two mirror sectors. Since $\sigma_{16} = K\beta$ with $\beta = -iT^4$, and K commutes with $T^0, \ldots, T^4$ (and anticommutes with T^5, T^6, T^7), every field that does not depend on x_5, x_6, x_7 splits as $\psi = \psi_+ + \psi_-$, $K\psi_\pm = \pm\psi_\pm$, and

$$\mathcal{L} = \mathcal{L}_D[\psi_+] - \mathcal{L}_D[\psi_-] - \frac{\lambda}{2}(S_+ - S_-)^2, \qquad \mathcal{L}_D[\chi] = \chi^\dagger\beta(T^A\partial_A - m)\chi \ (A = 0..4), \qquad S_\pm = \psi_\pm^\dagger\beta\psi_\pm.$$

The 16-component field is thus a pair of 8-component universes whose energies (masses) enter with opposite signs: a "+mass" and a "-mass" universe, coupled by $-\lambda S_+S_-$ inside $-\frac{\lambda}{2}(S_+ - S_-)^2 = -\frac{\lambda}{2}(S_+^2 + S_-^2) + \lambda S_+S_-$.

7. Mirroring Pre-gravity6plane.nb: dirac16complex depending on x_0 and x_4

Pre-gravity6plane.nb studies $\Psi_{16} = f16[k][x_0, x_4]$, a wave function of a Universe depending only on x_0 and x_4, with Lagrangian $L_a = \frac{1}{H}\Psi_{16}^T\sigma_{16}\sum_A T^A\partial_A\Psi_{16} + \frac{2M}{H}\Psi_{16}^T\sigma_{16}\Psi_{16}$ and field equation $T^A\partial_A\Psi_{16} + 2M\Psi_{16} = 0$. The corresponding configurations of dirac16complex are those that depend only on x_0 and x_4 (no dependence on $x_1, x_2, x_3, x_5, x_6, x_7$). For them the field equation is

$$T^0\partial_0\psi + T^4\partial_4\psi = (m + \lambda\bar{\psi}\psi)\psi, \qquad \text{with } m = -2M \text{ to match the old notebook.}$$

Couplings (analogue of eLaCouplings). The 16 equations split into four independent systems of four components:

$$\{0,5,8,13\},\quad \{1,4,9,12\},\quad \{2,7,10,15\},\quad \{3,6,11,14\},$$

exactly the pattern found in Pre-gravity6plane.nb.

Variables $z = 6Hx_0$, $t = Hx_4$ (analogue of eLazt). The linear equation becomes $6T^0\partial_z\psi + T^4\partial_t\psi = \frac{m}{H}\psi$. A mode $e^{pz+qt}u$ solves it iff $(6pT^0 + qT^4 - \frac{m}{H})u = 0$, possible iff

$$(6p)^2 - q^2 = \left(\frac{m}{H}\right)^2 \quad\Longleftrightarrow\quad (6p+q)(6p-q) = \frac{m^2}{H^2}.$$

Writing $6p + q = 2C$ and $m = -2M$, the exponent is

$$pz + qt = C\left(t + \frac{z}{6}\right) - \frac{M^2}{CH^2}\left(t - \frac{z}{6}\right),$$

which is exactly the family of exponential solutions that Maple produced in Pre-gravity6plane.nb ("Solution 15feb"); the field equation then reads $(6pT^0 + qT^4 + \frac{2M}{H})u = 0$ and the spinor is $u = (6pT^0 + qT^4 - \frac{2M}{H})w$ for any constant w, because $(6pT^0 + qT^4 + \frac{2M}{H})(6pT^0 + qT^4 - \frac{2M}{H}) = (36p^2 - q^2 - \frac{4M^2}{H^2})I_{16} = 0$. For a complex field the generic solutions are oscillatory plane waves:

$$\psi = e^{i(kx_0 - Ex_4)}\left(ikT^0 - iET^4 + m\right)w, \qquad E = \pm\sqrt{k^2 + m^2}.$$

Retarded and advanced variables. $z_{adv} = 6t + z$, $z_{ret} = 6t - z$ (equivalently $\xi_{adv} = x_0 - x_4$, $\xi_{ret} = x_0 + x_4$).

Light-like quanta. For $m = 0$ the solution $\psi = e^{ik(x_0-x_4)}\, ik(T^0 - T^4)w$ depends on one light-cone variable only. Because $(T^0 - T^4)^2 = (T^0)^2 + (T^4)^2 = 0$, every component of the current $J^A = i\bar\psi T^A\psi$ and the scalar $\bar\psi\psi$ vanish identically, just as the bilinears $\psi\sigma_{16}\psi$ vanished for the light-like quanta of Pre-gravity6plane.nb.

8. Triality frames and vector forms

The unit spinor is $u = \frac{1}{\sqrt{2}}(1,0,0,0,1,0,0,0)$, the 8th normalized eigenvector of σ ($u^T\sigma u = 1$). The triality frames of Pre-gravity6plane.nb are

$$(F_A{}^a)_{aA} = \eta_{AA}(\tau_A u)_a, \qquad (F_a{}^A)_{Aa} = (u^T\sigma\bar\tau_A)_a, \qquad F_A{}^a F_a{}^A = F_a{}^A F_A{}^a = I_8.$$

For a Universe ψ write $\psi_1 = (\psi_0,\dots,\psi_7)$ (type 1) and $\psi_2 = (\psi_8,\dots,\psi_{15})$ (type 2), and define the vector forms

$$V_2 = F_a{}^A\,\psi_2, \qquad V_1 = \eta\,(\psi_1^T\sigma F_A{}^a).$$

9. The 6-planes of dirac16complex

The field equation is real, so the real part of every Universe that depends only on x_0 and x_4 is itself a real solution of the type studied in Pre-gravity6plane.nb. Apply the old construction to it: $n_1 = V_1[\mathrm{Re}\,\psi]$ and $n_2 = V_2[\mathrm{Re}\,\psi]$ are real 8-vectors depending on (x_0, x_4), and with $Z = (z_1,\dots,z_8) \in \mathbb{R}^8$:

$$7\text{-plane}_k = \{Z : n_k \cdot Z = 0\}, \qquad M^6 = 7\text{-plane}_1 \cap 7\text{-plane}_2 = \{Z : n_1\cdot Z = 0,\ n_2\cdot Z = 0\}.$$

M^6 is 6-dimensional whenever n_1 and n_2 are linearly independent. Solving for a pair of coordinates $\{z_i, z_j\}$ is possible iff the minor $n_{1i}n_{2j} - n_{1j}n_{2i} \neq 0$; there are $\binom{8}{2} = 28$ such pairs, the 28 parameterizations of M^6. **Example** (computed in 6planes.nb): for $\psi = e^{i(\frac{3}{4}x_0 - \frac{5}{4}x_4)}(\frac{3}{4}iT^0 - \frac{5}{4}iT^4 + 1)w$ with $w = (1,2,\dots,16)$ at $(x_0, x_4) = (0.3, 0.7)$, the rank of $\{n_1, n_2\}$ is 2 and all 28 minors are non-zero, and

$$z_7 = -132.890\,z_1 + 189.925\,z_2 - 174.913\,z_3 + 155.902\,z_4 + 77.0462\,z_5 - 74.0462\,z_6,$$

$$z_8 = -149.525\,z_1 + 213.689\,z_2 - 196.301\,z_3 + 174.913\,z_4 + 87.5520\,z_5 - 84.5520\,z_6.$$

A pair of universes creates a 6-plane by itself. For a single plane wave, $V_2 = e^{i\phi}\times$(real vector), so $\mathrm{Re}\, V_2$ and $\mathrm{Im}\, V_2$ are parallel and only give a 7-plane. For the superposition of a positive-energy and a negative-energy Universe,

$$\psi_{pair} = e^{i(kx_0 - Ex_4)}(ikT^0 - iET^4 + m)w + e^{i(kx_0 + Ex_4)}(ikT^0 + iET^4 + m)w', \qquad E = \sqrt{k^2 + m^2},$$

$\mathrm{Re}\, V_2$ and $\mathrm{Im}\, V_2$ are independent (rank 2) and

$$M^6_{pair} = \{Z : \mathrm{Re}\, V_2 \cdot Z = 0,\ \mathrm{Im}\, V_2 \cdot Z = 0\}$$

is a 6-plane that moves with (x_0, x_4): the pair $\{+E, -E\}$ of universes generates a 6-dimensional plane.

10. What a student should remember

1. The Dirac matrices are real because the signature is (4,4); Cl(4,4) is the algebra of all real 16×16 matrices.
2. The field must be complex for $\bar{\psi}\psi$ and $(\bar{\psi}\psi)^2$ to be non-zero.
3. The field equation $T^A\partial_A\psi = (m + \lambda\bar{\psi}\psi)\psi$ follows from the Grassmann Euler-Lagrange equations.
4. With four times, quantization has an indefinite metric and the field splits into two mirror universes with opposite energies.
5. Universes (mode functions) produce 7-planes through their vector forms; two 7-planes, or a single $\pm$ pair of universes, produce a 6-plane M^6.